

C. Double Happiness

time limit per test: 3 seconds
memory limit per test: 128 megabytes

On the math lesson a teacher asked each pupil to come up with his own lucky numbers. As a fan of number theory Peter chose prime numbers. Bob was more original. He said that number t is his lucky number, if it can be represented as:

$$t = a^2 + b^2,$$

where a, b are arbitrary positive integers.
Now, the boys decided to find out how many days of the interval $[l, r]$ ($l \leq r$) are suitable for pair programming. They decided that the day i ($l \leq i \leq r$) is suitable for pair programming if and only if the number i is lucky for Peter and lucky for Bob at the same time. Help the boys to find the number of such days.

Input

The first line of the input contains integer numbers l, r ($1 \leq l, r \leq 3 \cdot 10^8$).

Output

In the only line print the number of days on the segment $[l, r]$, which are lucky for Peter and Bob at the same time.

Examples

input	Copy
3 5	
output	Copy
1	

input	Copy
6 66	
output	Copy
7	

Codeforces Beta Round 86 (Div. 1 Only)

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms)

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
361768408	Feb/07/2026 01:02	Accepted
361768356	Feb/07/2026 01:01	Time limit exceeded on test 1
361767937	Feb/07/2026 00:52	Time limit exceeded on test 1
361767770	Feb/07/2026 00:48	Wrong answer on test 30
361767605	Feb/07/2026 00:45	Time limit exceeded on test 1
361767492	Feb/07/2026 00:42	Time limit exceeded on test 1
361767254	Feb/07/2026 00:37	Time limit exceeded on test 1
361766855	Feb/07/2026 00:30	Accepted
361766729	Feb/07/2026 00:28	Time limit exceeded on test 11
361470767	Feb/05/2026 01:07	Accepted

→ Problem tags

brute force

math

number theory

*2200

No tag edit access

- Announcement ☐
- Tutorial ☐

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